

CS407 In Class Activity

Logistic Regression and kNN Classification

Your group will compare scaled logistic regression with scaled k-nearest neighbors classification. Each group receives a controlled synthetic variation. Use the common procedure so the class can compare how data conditions affect model behavior and evaluation.

Synthetic instructional data All records were generated for classroom use and do not represent actual students. Model associations do not establish that changing study habits, attendance, sleep, stress, or another predictor causes a change in academic risk. Treat the results as evidence within these synthetic samples.

Format and submission Use only your group's assigned file. The Group POC must submit one completed notebook and the group's presentation slides before the presentation. Keep the provided train and test assignments unchanged.

Learning goals

- Explain how logistic regression converts a linear score into a class probability.
- Explain how scaling and the number of neighbors affect kNN classification.
- Evaluate classifiers with threshold metrics, ranking metrics, and a majority-class baseline.
- Explain how a controlled data variation changes performance, interpretation, or generalization.

Dataset assignments

Group	Team	Dataset	Variation
1	PEM	synthetic_classification_group_1.csv	Test population shift
2	Team Terminal	synthetic_classification_group_2.csv	Minority positive class
3	OMAC	synthetic_classification_group_3.csv	Extreme numeric observations
4	SKA	synthetic_classification_group_4.csv	Irrelevant categorical predictors
5	Black Eagle	synthetic_classification_group_5.csv	Clean baseline

Tasks

- 1. Define the modeling problem** Use Risk as the binary target y . Risk=1 is the positive class and indicates that the synthetic student requires support. Exclude Synthetic_Record_ID and Split from every model.
- 2. Inspect the data and the fixed split** Report record and field counts, data types, missing values, duplicate identifiers, plausible ranges, class proportions, and unusual observations. Compare the train and test distributions without using the test outcomes to make model choices.
- 3. Prepare the predictors** Place preprocessing inside each pipeline. Standardize numeric predictors. Group 4 must one-hot encode its categorical predictors with `handle_unknown='ignore'`. Exclude the identifier and split columns.
- 4. Fit scaled logistic regression** Use `StandardScaler` followed by `LogisticRegression(max_iter=2000)`. Report standardized coefficients and odds ratios. Interpret each coefficient while holding the other modeled predictors constant.
- 5. Tune scaled kNN** Use `StandardScaler` followed by `KNeighborsClassifier`. Evaluate k values of 3, 5, 9, 15, and 25 with five-fold stratified validation on the training rows. Select k using mean balanced accuracy.
- 6. Evaluate and compare** Refit the final pipelines on all training rows, then evaluate the untouched test rows at a 0.50 threshold. Compare both models with the majority-class baseline using the required metrics and plots.
- 7. Interpret the variation** Complete the group-specific comparison, explain why the variation affected the two models differently, and state one limitation and one non-causal interpretation of an observed association.

Models to compare

Retain the four results below. The majority-class result establishes a minimum reference point. The fixed kNN model checks the effect of the common starting value, while the selected kNN model shows whether training-only tuning improves estimated generalization.

Model	Preprocessing	Purpose
Majority-class baseline	No predictors	Reference for class imbalance
Scaled logistic regression	Numeric scaling; categorical encoding when needed	Linear log-odds model
Scaled kNN, $k=5$	Numeric scaling; categorical encoding when needed	Common kNN starting point
Selected scaled kNN	Same pipeline with training-selected k	Final kNN comparison

Common evaluation procedure

Every group must use the settings below. The fixed procedure keeps preprocessing, tuning, thresholds, and test evaluation comparable across the five data variations.

Item	Required setting
Fixed split	Develop on the 750 train rows. Use the 250 test rows only after model choices are complete.
kNN tuning	StratifiedKfold with 5 splits, shuffle=True, random_state=407; k values 3, 5, 9, 15, and 25
Selection rule	Highest mean validation balanced accuracy; use ROC AUC and variability as supporting evidence
Test metrics	At threshold 0.50: accuracy, balanced accuracy, precision, recall, F1, ROC AUC, and average precision
Leakage rule	Fit encoding, scaling, and k selection inside the training workflow; never fit on test rows

Presentation and submission requirements

- One completed notebook and one concise slide deck submitted by the Group POC
- Target, positive-class, predictor, and dataset-variation definitions
- Basic data-quality findings, including train and test class proportions
- Majority baseline, scaled logistic regression, kNN with k=5, and selected kNN results
- A comparison table containing every required test metric
- Confusion matrices plus one ROC or precision-recall comparison plot
- Standardized logistic coefficients and held-out permutation importance

Group specific questions

Group	Question to answer
1	Which predictors shifted between train and test, and why did the shift affect logistic regression and kNN differently?
2	Why can accuracy look strong when Risk=1 is uncommon? How does class_weight='balanced' change logistic precision and recall?
3	Which values are implausible? How do StandardScaler and RobustScaler change kNN distances and test performance?
4	Do the two categorical predictors improve either model? What changes when they are removed?
5	Which relationships and model results provide the clean reference for the other groups?